

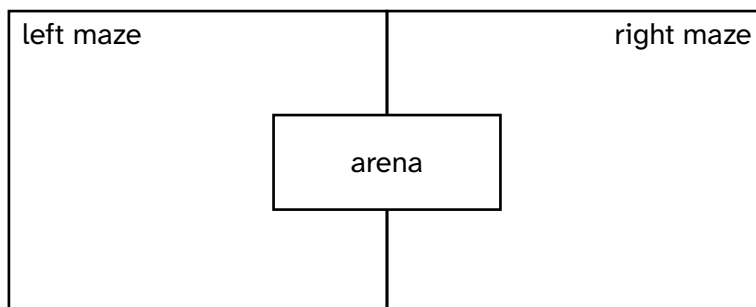
Dungeon Arena

Implement a terminal-based dungeon crawler where a human player races through a procedurally generated maze against an AI opponent. Both sides collect items to improve their stats and fight monsters to level up. One player can break through to the other side to hunt down their opponent.

The game ends when either the player or the opponent drops to 0 hit points.

Play Area

The play area fills the terminal window and is divided into three parts:



- The **left maze** is navigated by Player 1 (human-controlled).
- The **right maze** is navigated by Player 2 (AI-controlled).
- The **arena** sits in the center, separating the two halves. Each side has a door through which only the corresponding player may pass.

The last row of the terminal is reserved for a status bar showing each player's current level, hit points, attack, and defense.

Characters

Each character — players and monsters alike — has the following attributes:

- **Glyph and colour** used to draw them on screen.
- **Hit points (HP)** — reaching or falling below zero means death.
- **Attack and defense** — damage dealt per round is $\max(\text{attacker.atk} - \text{defender.def}, 1)$.
- **Level** (player only) — increases on a kill, granting +10 HP, +1 attack, and +1 defense.
- **Sense range** (monsters only) - the taxicab radius within which they can detect a player.

Maze Generation

Each half of the play area is a procedurally generated maze. After generation, a number of **rooms** are carved out by picking random center points and clearing all walls within a random Chebyshev radius. A small number of additional walls are also removed at random to increase connectivity.

Items

Weapons (⚔), and armor (🛡) are scattered randomly across each half. Walking onto an item equips it immediately, adding its bonus to attack or defense respectively. The number of items and the improvement they bestow is randomly generated, but each half receives the same number and quality of items.

Monsters

Different monster types inhabit each half of the maze. The following is one possible set, using NetHack glyphs as inspiration — feel free to invent your own:

Glyph	Name	HP	ATK	DEF	Sense
g	Gremlin	6	2	1	8

D	Dragon	20	6	3	12
G	Gnome	5	1	2	6
j	Jelly	8	3	0	10

Outside their sense range, monsters move randomly (always to a passable cell). Once a player enters their sense range, they switch to pathfinding towards that player.

Traps

Traps are hidden hazards scattered across both halves of the maze. They are invisible until a character steps on them, at which point they are sprung and become visible (shown as ^), but are spent and cannot trigger again.

Each trap randomly reduces one of the victim's stats:

- **HP** — the character loses hit points immediately; reaching zero is fatal.
- **ATK** — attack is reduced, to a minimum of 0.
- **DEF** — defense is reduced, to a minimum of 0.

Monsters killed by traps count as a kill for their side's player.

Combat

Combat is triggered by moving into a cell occupied by another character. Whoever initiates the move attacks first. Each round, the attacker deals $\max(\text{attack} - \text{defense}, 1)$ damage; if the defender survives, they retaliate with the same formula. Killing a monster grants the player one level-up.

The Arena Doors

Each arena wall has a coloured door, passable only by the corresponding player:

- **P1's door** (blue) is on the left arena wall.
- **P2's door** (green) is on the right arena wall.

The *first* player to pass through their door gains a bonus level-up. Passing through closes that door permanently (blocking retreat) and removes the opposite door entirely, opening it to all characters.

Player 2 AI

The AI controls Player 2 and operates in three sequential phases:

1. **Collect** — path to the nearest item on the right half using pathfinding.
2. **Hunt monsters** — once all items are collected, seek out the weakest remaining monster (by current HP) and fight it.
3. **Hunt Player 1** — once all monsters are dead, path through the arena door and pursue Player 1.

Win and Loss Conditions

- **Player 1 loses** if their HP reaches zero — a "YOU DIED" banner is shown.
- **Player 1 wins** if Player 2's HP reaches zero for any reason — a "YOU WIN" banner is shown.

Controls

W / A / S / D	Move Player 1 up / left / down / right
Q	Quit

Optional Extensions

- Visualise maze generation step by step
- Make monsters, items, traps, and maze generation parameters configurable.

- Use line-of-sight instead of radius for monster detection
- Make the game real-time instead of turn-based
- Support two human players

Links

- NetHack Guidebook <https://nethack.org/v500/Guidebook.html>
- ANSI Escape Codes https://en.wikipedia.org/wiki/ANSI_escape_code
- TIOCGWINSZ <https://www.man7.org/linux/man-pages/man2/TIOCGWINSZ.2const.html>
- Distance Metrics
 - Chebyshev Distance https://en.wikipedia.org/wiki/Chebyshev_distance
 - Taxicab Geometry https://en.wikipedia.org/wiki/Taxicab_geometry
- Maze Generation
 - Wilson's Algorithm <https://weblog.jamisbuck.org/2011/1/20/maze-generation-wilson-s-algorithm>
 - Aldous-Broder <https://weblog.jamisbuck.org/2011/1/17/maze-generation-aldous-broder-algorithm>
- Introduction to A* <https://www.redblobgames.com/pathfinding/a-star/introduction.html>